Lesson notes · for you

Training and testing

Data science · Level 3 · Lesson 2

Time
30 min

Materials
A notebook and a pencil; 15 small cards or slips of paper; A phone or tablet with Darsi, for the lab

Learning goals

- Split data to train and test.
- Read training and test errors.
- Spot overfitting.

Activity: Hidden cards

1. Warm-up (5 min): Say whether knowing the exact exam questions would show what you learned.
2. Build (10 min): With a partner, invent a rule from ten cards and work out its training error.
3. Test (10 min): Turn over the hidden cards and work out the test error.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Did your rule work on the hidden cards?
- Why not peek at the test cards?
- What is overfitting?

Check yourself

- In one sentence, why do we keep a test set?

Try it this week
Make up a rule to guess a friend's mark from their homework hours using ten cards, then test it on five hidden cards. Did your rule still work?

Worksheet

Training and testing

Data science · Level 3 · Lesson 2

Name _____ Class _____ Date _____

1 **Why do we keep 20% of the data away from the model while we build it?**

◯ To make it faster ◯ To check it on pupils it has never seen ◯ Because that data is wrong

2 **Match each result to its diagnosis.**

Train error 1, test error 9	• • • •	A good fit
Train error 8, test error 9	• • • •	Overfitting
Train error 3, test error 3.5	• • • •	Underfitting

3 **The rule predicts 8 + 2 * x for each row. Split 8 rows for training and 2 for testing, print both sizes, then print the rule's mean absolute error on the 2 test rows.**

```
x = [0, 1, 1, 2, 3, 3, 4, 5,
     2, 4]
y = [8, 10, 11, 12, 13, 15, 16, 18,
     15, 13]
train, test = x, []
print(len(train), len(test))
```

Write your program:

4 **Tuning a model again and again on the test set keeps the test score honest.**

◯ True ◯ False

5 **Put the workflow in order.**

☐ Fit on the training set

☐ Report the test score

☐ Split the data

☐ Evaluate once on the test set

✎ **Have you ever learned answers by heart and then struggled with a new question? How is that like overfitting?**

Check your answers

Training and testing

Data science · Level 3 · Lesson 2

1 **To check it on pupils it has never seen**

Held-back data gives an honest score.

2 **Train error 1, test error 9 — Overfitting**

Train error 8, test error 9 — Underfitting

Train error 3, test error 3.5 — A good fit

A big gap means overfitting; two high errors mean underfitting.

3

```
x = [0, 1, 1, 2, 3, 3, 4, 5,
     2, 4]
y = [8, 10, 11, 12, 13, 15, 16, 18,
     15, 13]
train, test = x[:8], x[8:]
print(len(train), len(test))
e = 0
for i in range(8, 10):
    e += abs(y[i] - (8 + 2 * x[i]))
print(e / 2)
```

Both test rows miss by 3, so the test error is 3.0, far worse than on the training rows.

4 **False**

The test set leaks into training and the score becomes optimistic.

5 **Top to bottom: 2, 4, 1, 3**

Split first so the test set stays unseen.